

- Change-of-coefficient homomorphisms $H_n(X; G_1) \rightarrow H_n(X; G_2)$ induced by a homomorphism $G_1 \rightarrow G_2$, as in the proof of Lemma 2.49.

In general, if one has two functors $F, G: \mathcal{C} \rightarrow \mathcal{D}$ then a **natural transformation** T from F to G assigns a morphism $T_X: F(X) \rightarrow G(X)$ to each object $X \in \mathcal{C}$, in such a way that for each morphism $f: X \rightarrow Y$ in $\mathcal{C}$ the square at the right commutes. The case that F and G are contravariant rather than covariant is similar.

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ \downarrow T_X & & \downarrow T_Y \\ G(X) & \xrightarrow{G(f)} & G(Y) \end{array}$$

We have been describing the passage from topology to the abstract world of categories and functors, but there is also a nice path in the opposite direction:

- To each category $\mathcal{C}$ there is associated a Δ -complex $B\mathcal{C}$ called the **classifying space** of $\mathcal{C}$, whose n -simplices are the strings $X_0 \rightarrow X_1 \rightarrow \cdots \rightarrow X_n$ of morphisms in $\mathcal{C}$. The faces of this simplex are obtained by deleting an X_i , and then composing the two adjacent morphisms if $i \neq 0, n$. Thus when $n = 2$ the three faces of $X_0 \rightarrow X_1 \rightarrow X_2$ are $X_0 \rightarrow X_1$, $X_1 \rightarrow X_2$, and the composed morphism $X_0 \rightarrow X_2$. In case $\mathcal{C}$ has a single object and the morphisms of $\mathcal{C}$ form a group G , then $B\mathcal{C}$ is the same as the Δ -complex BG constructed in Example 1B.7, a $K(G, 1)$. In general, the space $B\mathcal{C}$ need not be a $K(G, 1)$, however. For example, if we start with a Δ -complex X and regard its set of simplices as a partially ordered set $\mathcal{C}(X)$ under the relation of inclusion of faces, then $B\mathcal{C}(X)$ is the barycentric subdivision of X .
- A functor $F: \mathcal{C} \rightarrow \mathcal{D}$ induces a map $B\mathcal{C} \rightarrow B\mathcal{D}$. This is the Δ -map that sends an n -simplex $X_0 \rightarrow X_1 \rightarrow \cdots \rightarrow X_n$ to the n -simplex $F(X_0) \rightarrow F(X_1) \rightarrow \cdots \rightarrow F(X_n)$.
- A natural transformation from a functor F to a functor G induces a homotopy between the induced maps of classifying spaces. We leave this for the reader to make explicit, using the subdivision of $\Delta^n \times I$ into $(n + 1)$ -simplices described earlier in the chapter.

Exercises

1. If $T_n(X, A)$ denotes the torsion subgroup of $H_n(X, A; \mathbb{Z})$, show that the functors $(X, A) \mapsto T_n(X, A)$, with the obvious induced homomorphisms $T_n(X, A) \rightarrow T_n(Y, B)$ and boundary maps $T_n(X, A) \rightarrow T_{n-1}(A)$, do not define a homology theory. Do the same for the 'mod torsion' functor $MT_n(X, A) = H_n(X, A; \mathbb{Z}) / T_n(X, A)$.
2. Define a candidate for a reduced homology theory on CW complexes by $\tilde{h}_n(X) = \prod_i \tilde{H}_i(X) / \bigoplus_i \tilde{H}_i(X)$. Thus $\tilde{h}_n(X)$ is independent of n and is zero if X is finite-dimensional, but is not identically zero, for example for $X = \bigvee_i S^i$. Show that the axioms for a homology theory are satisfied except that the wedge axiom fails.
3. Show that if $\tilde{h}$ is a reduced homology theory, then $\tilde{h}_n(\text{point}) = 0$ for all n . Deduce that there are suspension isomorphisms $\tilde{h}_n(X) \approx \tilde{h}_{n+1}(SX)$ for all n .
4. Show that the wedge axiom for homology theories follows from the other axioms in the case of finite wedge sums.

Additional Topics

2.A Homology and Fundamental Group

There is a close connection between $H_1(X)$ and $\pi_1(X)$, arising from the fact that a map $f:I \rightarrow X$ can be viewed as either a path or a singular 1-simplex. If f is a loop, with $f(0) = f(1)$, this singular 1-simplex is a cycle since $\partial f = f(1) - f(0)$.

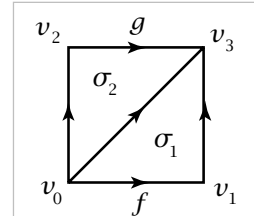
Theorem 2A.1. *By regarding loops as singular 1-cycles, we obtain a homomorphism $h:\pi_1(X, x_0) \rightarrow H_1(X)$. If X is path-connected, then h is surjective and has kernel the commutator subgroup of $\pi_1(X)$, so h induces an isomorphism from the abelianization of $\pi_1(X)$ onto $H_1(X)$.*

Proof: Recall the notation $f \simeq g$ for the relation of homotopy, fixing endpoints, between paths f and g . Regarding f and g as chains, the notation $f \sim g$ will mean that f is homologous to g , that is, $f - g$ is the boundary of some 2-chain. Here are some facts about this relation.

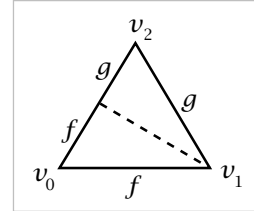
(i) If f is a constant path, then $f \sim 0$. Namely, f is a cycle since it is a loop, and since $H_1(\text{point}) = 0$, f must then be a boundary. Explicitly, f is the boundary of the constant singular 2-simplex σ having the same image as f since

$$\partial\sigma = \sigma|_{[v_1, v_2]} - \sigma|_{[v_0, v_2]} + \sigma|_{[v_0, v_1]} = f - f + f = f$$

(ii) If $f \simeq g$ then $f \sim g$. To see this, consider a homotopy $F:I \times I \rightarrow X$ from f to g . This yields a pair of singular 2-simplices σ_1 and σ_2 in X by subdividing the square $I \times I$ into two triangles $[v_0, v_1, v_3]$ and $[v_0, v_2, v_3]$ as shown in the figure. When one computes $\partial(\sigma_1 - \sigma_2)$, the two restrictions of F to the diagonal of the square cancel, leaving $f - g$ together with two constant singular 1-simplices from the left and right edges of the square. By (i) these are boundaries, so $f - g$ is also a boundary.



(iii) $f \cdot g \sim f + g$, where $f \cdot g$ denotes the product of the paths f and g . For if $\sigma:\Delta^2 \rightarrow X$ is the composition of orthogonal projection of $\Delta^2 = [v_0, v_1, v_2]$ onto the edge $[v_0, v_2]$ followed by $f \cdot g:[v_0, v_2] \rightarrow X$, then $\partial\sigma = g - f \cdot g + f$.



(iv) $\bar{f} \sim -f$, where $\bar{f}$ is the inverse path of f . This follows from the preceding three observations, which give $f + \bar{f} \sim f \cdot \bar{f} \sim 0$.

Applying (ii) and (iii) to loops, it follows that we have a well-defined homomorphism $h:\pi_1(X, x_0) \rightarrow H_1(X)$ sending the homotopy class of a loop f to the homology class of the 1-cycle f .